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Digital Government Interoperability and Data Exchange Platforms: Insights from a Twenty Country Comparative Study

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A core component for building a digital government is a governmental interoperability and data exchange platform. To be a successful digital government interoperability and the ability to exchange data are a must. However, the research into such government interoperability and data exchange platforms, their conceptualization, structure, and core components is lacking. This paper aims to make an initial contribution to the research and create new knowledge on these platforms by conducting a comparative twenty country case study that focuses specifically on the architecture, design, and governance of government interoperability and data exchange platforms. As a result of this research, it is possible to offer a new conceptualization and theorization of these platforms, offer new insights into their structure, form, and governance, and make initial steps towards identifying critical building blocks for the future of government interoperability and data exchange platforms.

Additional Keywords and Phrases: Interoperability, Digital Government, Data Exchange, Platform, Innovation, eGovernment

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1 INTRODUCTION

Without the presence of interoperability and the ability to exchange data, it is not possible to have a functioning digital government. Interoperability is essential. At its simplest, (technical) interoperability describes the ability of two systems to exchange data with one another, understand this data, and use it. Interoperability can also be defined on a number of other aspects such as the organizational, legal, or political components. In this paper, technical interoperability is the primary object of interest. Within the context of digital government, interoperability is what enables digital government to function by allowing for the exchange of data between agencies, organizations, services, and other actors.

Due to the importance of interoperability for the digitalization of government, investments into government interoperability and data exchange platforms (GIDEPs) are growing. A GIDEP can be understood as the technical infrastructure and architecture that facilitates the exchange of data between organizations. Examples of such platforms include the proposed GAIA-X within the EU or Estonia's X-Road. Though GIDEPs are important, there is a lack of up-to-date, empirical, and (especially) comparative research focusing on them. Thus, a situation emerges where many resources are being devoted to developing new GIDEPs or innovating already existing systems, but there is little work theorizing or conceptualizing such platforms. Additionally, there is little academic understanding about the structure of such platforms, the state-of-the-art, and the potential future for GIDEPs. Thus, there is a clear need for empirical and comparative research on the topic of GIDEPs that is able to take stock of the state-of-the-art and conceptualize the developments of such platforms. This paper addresses this need.

The research for this paper starts by conceptualizing and theorizing GIDEPs from the perspective of digital infrastructure and digital innovation, develops new understanding about the structure, design, and core components of existing GIDEPs and, then, turns toward the future, outlining what GIDEPs of the future may look like. To reach these goals, the paper asks and answers four primary research questions. First, what are the common structures and design patterns for GIDEPs? Second, how are GIDEPs governed? Third, what are the most commonly

occurring building blocks within GIDEPs? Finally, fourth, what are the most important components for building a new state-of-the-art GIDEP?

To answer these questions, a twenty country (Australia, Belgium, Canada, Croatia, Denmark; Greece, Hungary, Israel, Korea, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, United Kingdom, Uruguay, and the United States) comparative case study was conducted with a specific focus on the architecture, design, and governance of GIDEPs. As a result of this research, it is possible to generate a new understanding about how GIDEPs functions, what the core technical building blocks and components are, how they are governed, and, ultimately, make initial steps towards forecasting and understanding the future of interoperability and data exchange within the public sector.

This research has important implications for the future of digital government, especially regarding topics such as innovation, citizen centricity, and overall government effectiveness. As this study represents one of the largest and most comprehensive studies on this topic to-date, it will be of interest for academics working on the field of interoperability and data exchange as well as digital government more broadly, for those within the public sector working on the topic of interoperability, as well as for those directly involved in the development of GIDEPs.

2 BACKGROUND

At the earliest stages of digital government research (at the time still called e-Government), there was a large emphasis on understanding interoperability and the exchange of data as an important "end-goal" of governmental digitalization [1]. To reach higher levels of interoperability, numerous countries developed their own e-GIF (eGovernment Interoperability Framework) plans [2]–[4] that outlined how their technology systems could become interoperable. Research focusing on interoperability and data exchange tended to agree that interoperability is challenging within the public sector, that it is extremely important for the development of a successful digitalized government, and that interoperability is a socio-technical phenomenon [5]–[11]. A large amount of earlier research approached interoperability and data exchange from normative, philosophical, or conceptual lenses, often lacking in strong empirical support. Emerging research still often adopts the same approach, but also tries to understand the more technical aspects of interoperability and data exchange – often done by looking at the data exchange(s) of one or two countries. For example [12], compare the data exchange systems of Netherlands and Estonia from an institutional design perspective, [13] explores the Greek data exchange system, [14] explores the data exchange system in Croatia, or [15]'s exploration in Uruguay. In [16], the authors include more cases, but are rather concerned with the overall governance of data exchange, rather than the technical design or structure of the existing GIDEPs. From a technical perspective, a recent paper by [17] explores the literature to understand core building blocks that exist within digital government architectures, one of which happens to be data exchange and interoperability. Though this previous research on interoperability and its importance for the digitalization of government exists, more effort is needed to properly conceptualize and theorize GIDEPs, especially from a technical perspective.

In this paper, GIDEPs are conceptualized as a form of digital infrastructure. There are several definitions for digital infrastructures. They have been defined as "standard-setting digital technologies enabling, constraining, and coordinating numerous actors' actions and interactions in ecosystems, field, or industries" [18]. A second definition similarly views digital infrastructures as "the basic information technologies and organizational structures, along with the related services and facilities necessary for an enterprise or industry to function" [19]. They have also been defined as "the computing and network resources that allow multiple stakeholders to orchestrate their service and content needs", examples of which include the Internet and smart phones [20]. Furthermore, digital infrastructures

have a number of unique features such as flexibility, scalability, or recursivity that separate them from traditional infrastructures [19]. Digital infrastructures are what form the backbone of any digitalization process, therefore understanding "digital infrastructures and their effects are becoming fundamental in understanding the effects of digitalization" [19]. Though there are numerous reasons such infrastructures are important for the digitalization process, for this paper, the two most important to highlight are interoperability and generativity.

As technical interoperability has already been defined in the paper's introduction, it will only be addressed briefly here. In short, for a digital infrastructure to grow, it must be interoperable. Interoperability itself is a complex issue that requires efforts to develop standards and engage numerous stakeholders [21]. From a technical perspective, two 'interoperate' systems must be able to communicate with each other using understandable protocols and standards, and the exchanged data must be clearly defined and accompanied by the appropriate form, standards, or metadata so that it can be understood. This leads to the second important aspect of digital infrastructures, generativity. Generativity is a concept initially theorized by [22] and can be understood as "the ability of users to generate new, valuable uses that are easy to distribute and are in turn sources of further innovation". When a digital infrastructure makes itself available and interoperable, anyone can access its data or resources, build services, drive innovation, and, ultimately, continue to grow the digital ecosystem surrounding this ecosystem. These ecosystems can lead to higher levels of efficiency, quality, and effectiveness.

Though much of the research existing on digital infrastructures focuses on private sector organizations, the public sector plays an especially important role when it comes to digital infrastructures as "governments are often in the best position and most able to promote or directly facilitate other facets of interoperability and thereby contribute to the health of the interoperability ecosystem generally" [9]. Returning to the concept of GIDEPs and viewing it from a digital infrastructure lens, several important ideas emerge. The first is that GIDEPs play an important role in driving broader digitalization and digital transformation. The second is that interoperability is an extremely important aspect in driving the growth of such platforms. The third is that GIDEPs have the potential to drive innovation, should such platforms become more interoperable and allow their services and data to be accessed by numerous stakeholder groups. Therefore, understanding GIDEPs as a form of digital infrastructure is paramount for any research looking into the future of digital government.

To offer an initial conceptualization of GIDEPs, this paper has adopted the following understanding:

> **GIDEPs are a specific form of digital infrastructure necessary for the digitalization of the public sector that consist of a government's technical infrastructure, services, and data that, through technical interoperability, enable the exchange of data, provision of services, and digital innovation.**

3 METHODOLOGY

This research started with the observation that there is little research exploring, especially comparatively, GIDEPs. This is even more so the case when it comes to research that explores the technical nature of such systems. This is surprising for three primary reasons. First, without interoperability and data exchange, there can be no digital government. Second, technology has advanced; the technologies that built the first digital government data exchange systems in the early 2000s are becoming increasingly outdated. Third, GIDEPs have the potential to drive the creation of new innovation ecosystems, both within and outside a government's borders. From this perspective, there is a clear need for research that explores, especially comparatively, the technical nature of government data exchange systems.

Thus, methodologically, this research was conducted as an inductive and exploratory multi-country comparative case study [23]. For this research twenty countries were selected that represent a wide range of geographic locations, government types, administrative traditions, government quality, size, and digital government rankings. By selecting a varied group of countries it was hoped to provide a stronger sample for the analysis and strengthen the external validity and generalizability of the findings. An overview of the countries selected, and the relevant contextual data, is provided below as Table 1.

Table 1 Overview of countries studied. Source: Authors.

Country	Government Type	Population (thousands) 2019	UNEGDI 2020	Waseda 2021	Quality of Governance (2021)	Region
Australia	Federal	25687.04	5	21	1.62	Oceania
Belgium	Federal	11556	41	24	1.12	Western Europe
Canada	Federal	38005.24	28	5	1.64	North America
Croatia	Unitary, Centralized	4047.2	51	NA	0.44	Southern Europe
Denmark	Unitary, Decentralized	5831.4	1	1	1.89	Northern Europe
Greece	Unitary, Centralized	10715.55	42	NA	0.44	Southern Europe
Hungary	Unitary, Decentralized	9749.76	52	NA	0.58	Eastern Europe
Israel	Unitary	9216.9	30	40	1.1	Western Asia
Korea	Unitary	51780.58	2	8	1.42	Eastern Asia
Netherlands	Unitary, Decentralized	17441.14	10	14	1.85	Western Euurope
New Zealand	Unitary	5084.3	8	7	1.59	Oceania
Norway	Unitary, Dencentralized	5379.48	13	18	1.94	Northern Europe
Portugal	Unitary, Centralized	10305.56	35	36	1.02	Southern Europe
Singapore	Unitary, Centralized	5685.81	11	2	2.34	South-Eastern Asia
Spain	Unitary, Regional	47451.57	17	27	0.89	Southern Europe
Sweden	Unitary, Decentralized	10353.44	6	12	1.72	Northern Europe
Switzerland	Federal	8636.9	16	15	2.02	Western Europe
United Kingdom	Unitary, Regional	67215.29	7	3	1.38	Northern Europe
United States	Federal	329484.12	9	4	1.32	North America
Uruguay	Unitary, Decentralized	3473.73	26	35	0.78	South America

The study utilized a wide variety of empirical sources of evidence such as: emails and communications with direct participants in the design or implementation of the data exchange ecosystem within the specific country, conference presentations, academic papers describing the data exchange systems within the specific country, government websites, internal presentations, procurement documentation, system technical descriptions and architectural documents, reports compiled for international organizations, policy reports, and other forms of grey literature. On average, more than five different sources of data were utilized to generate an understanding about the technical structure of each country's data exchange system, or its lack thereof. The use of a wide variety of sources is especially important for enabling the triangulation of research findings. To analyze the data, an inductive approach was utilized, guided by the research questions, and following a grounded theory approach [24]–[26].

For each country analyzed the aim of the research was to understand who was responsible for developing and maintaining the system, what the core technical building blocks were, the functions and capabilities of the data exchange system, the nature of data exchange (e.g. G2G, G2C, G2B, or any mix of the three), services offered or available through the system, and an understanding of how data is stored. The availability of data for each of the countries varied in terms of granularity. For some countries it has been possible to obtain the exact software and components used in the system, in others it has only been possible to gather a surface level understanding due to confidentiality concerns, and, in others, systems were either still being developed or did not appear to exist or be widely used. An overview of the sources of evidence used, by country, for this research is shown below as Table 2.

Table 2 Empirical evidence sources. Source: Authors.

Country	Emails/Direct Communications	Conference presentations	Acadeamic papers	Government websites	Internal presentations	Procurement documentation	System technical descriptions and architectural documents	Grey literature	Policy reports
Australia				X				X	
Belgium	X	X		X		X	X	X	X
Canada		X		X	X	X		X	
Croatia			X	X		X		X	X
Denmark		X		X	X		X	X	X
Greece	X		X	X		X		X	X
Hungary		X		X			X	X	X
Israel		X		X		X		X	
Korea		X	X	X				X	X
Netherlands		X	X	X	X		X	X	X
New Zealand	X	X		X	X			X	
Norway		X		X			X	X	X
Portugal	X			X	X			X	X
Singapore	X	X		X			X	X	
Spain		X	X	X	X			X	X

Sweden				X				X	X
Switzerland		X		X	X		X	X	X
United Kingdom	X			X				X	X
United States	X			X				X	
Uruguay	X		X	X	X		X	X	

4 RESULTS

This section provides the initial results and answers to the research questions that have emerged from the conducted study, the implications of such results will be further drawn out in the following discussion section.

The first research question of this paper asked what are the common structures and design patterns for GIDEPs? In analyzing the GIDEPs of the selected countries, three differentiable forms began to emerge. These forms can be classified based on their method of data exchange as well as their method of data storage, thereby creating a matrix, as shown in Figure 1.

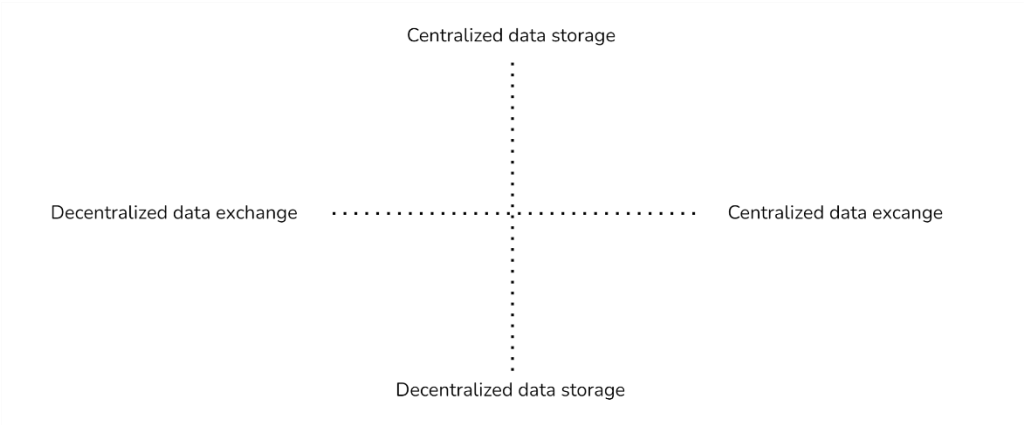

Figure 1 Government Interoperability and Data Exchange Matrix. Source: Authors.

The most commonly occurring form identified was to have a centralized data exchange with decentralized data storage (12 out of 20 countries studied), this was followed by those with both a decentralized data exchange and data storage (2 out of 20), the least common was to have a centralized data exchange and data storage (1 out of 20), in five countries studied there was no clearly identifiable data exchange and interoperability ecosystem. Example configurations of these interoperability and data exchange ecosystems are shown below as Figure 2.

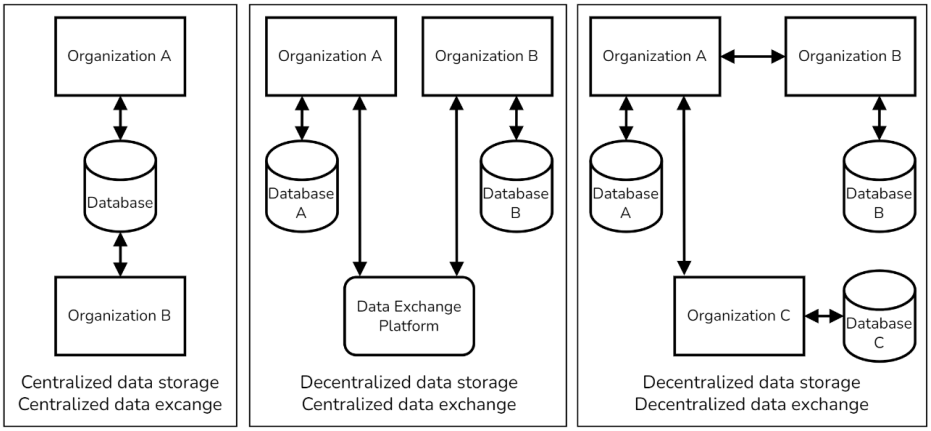

Figure 2: Government Interoperability and Data Exchange Configurations. Source: Authors.

In the United Kingdom, United States, Australia, New Zealand, and Sweden no GIDEPs were detectable. However, in all five countries there was a clear desire to develop their interoperability and data exchange capacity, primarily by developing an API data exchange ecosystem. This was often supported by API registries and API guidelines, enabling consistent API design and easier access to APIs and their descriptions. Should such an API based GIDEP emerge, it would likely be classified as having a decentralized data storage and a decentralized data exchange; an example of Sweden's conceptualization of this ecosystem is shown below as Figure 3.

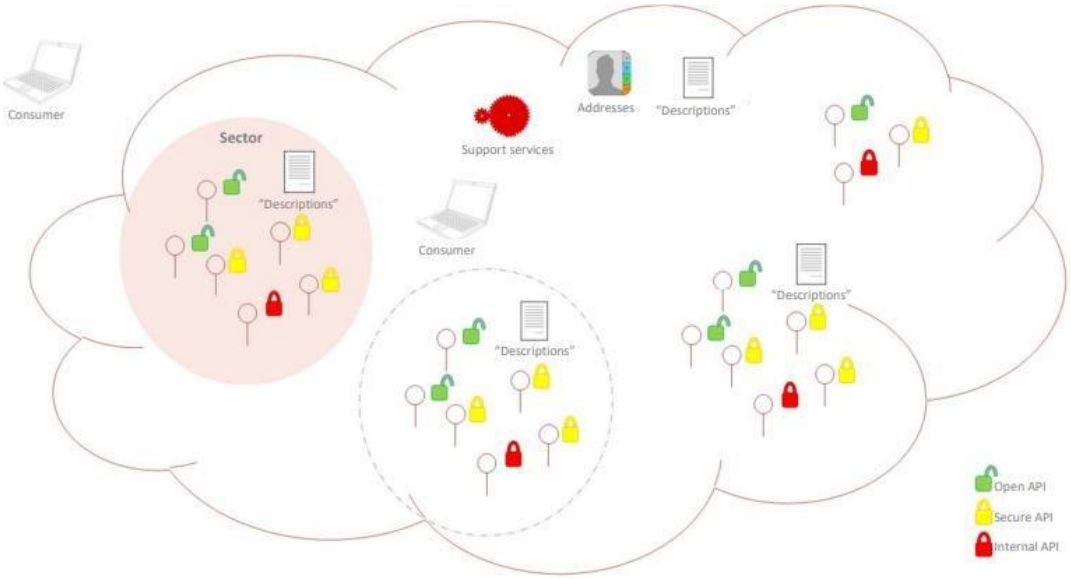

Figure 3 Sweden's Conceptualized API-Based GIDEP. Source: [27]

To answer the second research question, it was necessary to explore the relevant legal frameworks for each GIDEP. In most countries studied, a specific legal requirement to exchange data using their government interoperability and data exchange platform existed. However, there was significant deviation between the countries studied on the extent that these legal requirements existed and how they were formulated. One of the clearest areas of differentiation can be seen between unitary and federal countries. While in many of the unitary countries it was possible to mandate that government agencies use a specific interoperability platform (e.g. in Portugal), in federal countries legal agreements and cooperation agreements were first signed to cooperate and work together on interoperability and data exchange (as is the case in Belgium, Switzerland, and Spain). There are several other areas where there is significant divergence between countries when it comes to the legal frameworks relating to data exchange, particularly in areas related to data privacy, the once only principle, data standards, and to which organizations and levels of government the mandate applies to. To differentiate and classify each of the studied legal frameworks is beyond the scope of this current paper. However, a few commonly occurring fundamental components can be highlighted. In most cases these legal frameworks specified a single interoperability platform that must be used for the exchange of data, the standards and technologies associated with exchanging data on this platform are mandated, citizen consent for the exchange of personal information must be sought and recorded, and, increasingly, citizens should not be asked for their data more than once (the once only principle).

The studied GIDEPs were almost entirely the responsibility of a specific agency, this is inline with previous findings by [16]. These agencies tended to fill one of three roles: 1) they are responsible for digital government more broadly (e.g. in Norway); 2) they are responsible specifically for interoperability and data exchange (e.g. in Greece); or 3) they are responsible for government technology infrastructure more broadly (e.g. in Canada). These agencies themselves are often based within either a strategic Ministry (Finance or Interior) or within the Prime Minister's or President's Office. During this research, countries where the agency responsible for managing the interoperability and data exchange platform was focused on digital government broadly appeared to be more technologically advanced when compared to the other governance configurations.

In almost all countries studied, the interoperability and data exchange systems were exclusively for G2G data exchange, with a G2C component taking place only via a citizen portal and through various digital services. This is a surprising finding as the existence of GIDEPs should, in theory, help to open and drive innovation, but only if other stakeholder groups are able to join and access such information. The reason why GIDEPs currently are unable to fully reach their innovative potential appears to be due to legal framework compliance, security, or for technical reasons. Similarly, in most countries where a GIDEP was not detectable (Australia, New Zealand, USA, United Kingdom) the primary reason mentioned was due to the lack of a legal framework that enabled data exchange.

Though most GIDEPs were closed to organizations outside of the public sector, movement towards opening up these interoperability and data exchange systems was detectable. In South Korea new trials have begun to implement a MyData module [28] where citizens can send their data held in government sources to financial institutions [29] and, in Singapore, an API has been created for their digital identity platform that enables service providers to access information held by the government on the service user [30]. Similarly, in the Netherlands, one area where the government interoperability and data exchange systems is more openly available to nongovernmental users is the tax system [12].

The third research question for this paper asked what are the most commonly occurring building blocks within GIDEPs? At a very high level, there were a number of commonly occurring building blocks identified. The first group of building blocks are those that appear to be essential for any fully functional GIDEP: identity management, trust

(which refers to electronic services that help parties make binding decisions), digital services, data management, and secure data exchange. These four aspects were present in every GIDEP, they make take different forms, but they appear to be essential. Building blocks that were detectable in some, but not all, GIDEPs include those related to statistics and monitoring of the platforms, event-based processing, payment processing, messaging and communications, and data analytics.

5 DISCUSSION

It is already well established that when it comes to digitalizing the public sector, interoperability is foundational. Without interoperability and data exchange it will not be possible to build a digital government. At the earliest stages of the movement towards an "e-Government", interoperability and data exchange simply meant having the ability to move data across silos. Since these earliest developments, however, we have seen a slow movement on what this paper calls the continuum of interoperability, which is shown below as Figure 4, and ranges from no data exchange to a fully functional interoperable ecosystem. At the left side of the continuum, earlier advances in interoperability and data exchange are represented, for example when data is exchanged between two agencies under the same organization. Moving towards the right side of the continuum you start to see changes in how data are exchanged and interoperability is made available, with the end goal being a fully interoperable ecosystem, with data exchanged across borders and accessible to different stakeholder groups. Looking back to the background research, and approaching GIDEPs from a digital infrastructure perspective, this continuum also could be understood to represent generativity, with a GIDEP ecosystem having a higher level of generativity and innovation potential the further right one moves along the continuum.

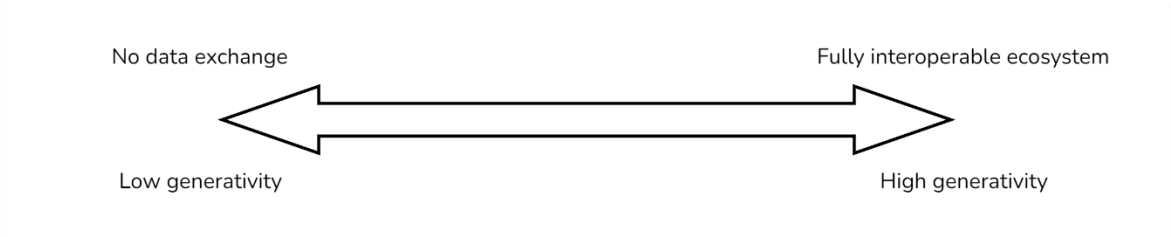

Figure 4 Continuum of Interoperability. Source: Authors.

In order to develop GIDEPs with a higher level of generativity and innovation potential, it is essential to understand what building blocks could play an influential role in opening up these platforms. It is precisely this need for understanding that sits at the heart of the fourth research question for this paper: what are the most important components for building a new state-of-the-art GIDEP? In this study, twenty countries had their GIDEPs examined and, by drawing on these insights, it is possible to begin to paint a picture about what the future may look like, and what core components are necessary to get there.

From a technical perspective, a number of clear changes appear when comparing countries with more advanced (or developing) GIDEPs with those that are less advanced or stagnated. When comparing GIDEPs across the countries included within the study, in the more advanced countries (Norway, Singapore, South Korea, Belgium, Portugal) there was a clear movement toward cloud-based hosting solutions for interoperability and data exchange systems and, in line with this, a movement towards open-source development as well as modular and microservice oriented architectures. Clear examples of this open source and modular development approach can be seen in Korea

with their eGovFrame program [31], which provides an open source and reusable architecture for building digital government services and in Norway's recently rebuilt Altinn system, which is cloud based, open source, and utilizes a microservice architecture [32]. These developments represent a change from the earliest approach to GIDEPs that tended to be closed source, enterprise based, and self-hosted. In today's technological environment, the earliest built GIDEPs have quickly become legacy infrastructure, leading to slow and inefficient systems with low potential for future innovation and development and a fairly high cost of maintenance. Changes can also be observed in how data is exchanged. While, traditionally, government interoperability and data exchange systems relied on web services and SOAP, there appears to be a movement towards systems based around APIs and REST. This is seen both in fully functional systems such as in Belgium, Norway, Singapore and Hungary, but also in systems currently being developed such as in Sweden, Uruguay, Israel, and Greece. The movement towards an API-based GIDEP is also often accompanied by the development of API standards and best practices, API registries, and an API-first based approached to digital service development

In line with these aforementioned innovations in GIDEPs, how data is used and processed within them is also being innovated upon. There are three clearly identifiable innovations that are likely to become increasingly important in interoperability and data exchange systems of the future: MyData, consent management, and the once only principle. MyData is a concept that envisages handing back control over data to citizens, creating new opportunities for individuals to exercise control over the data and how it is used [33]. MyData modules allow citizens to give permission to governmental or private sector stakeholders to use their data for service delivery. In line with the MyData movement is the presence of consent registers and consent management. In many countries included within this study the consent of the user is required to exchange data about them with another government organization. This creates the necessity of a consent register (Uruguay and Spain have great examples of such a register), where a citizen is able to view and access where and how their data is being processed, and the relevant consents they have or have not provided. While some examples of this do exist already, they rarely have a citizen focused perspective that allows them to view their provided consents. A third developing component related to data is that of the once only principle. From a technical perspective, the implementation of this principle requires strong mapping of data resources amongst government organizations and, additionally, must contain modules that allow for real time data querying and updates. From the countries included within this study, the best examples of this principle can be seen in Denmark, the Netherlands, and Belgium.

A final aspect to discuss is related to the relationship between a country's structure and governance type, and the structure and organization of their GIDEP. This relationship is especially clear when comparing the structure of platforms in unitary states and in the structure of platforms in federal states. In the federal states interoperability requires higher levels of cooperation and collaboration amongst stakeholders to drive interoperability. This makes the development of GIDEPs in federal states harder than in unitary states. Similarly, the GIDEPs in federal states appear to be more decentralized and like the structure of their country's governance. This is especially clear in Belgium and Spain. In these countries, the GIDEPs are multi-level, have specific roles for different levels of government, and are technically more complicated. On the other hand, in unitary states, the pattern holds. The clearest example coming from South Korea, which has a centralized mechanism for data exchange and data storage. This has important implications and seems to demonstrate that a country's digital government systems will recreate already existing practices and norms, this is a clear demonstration of Conway's law [34].

6 CONCLUSION

This paper started from the understanding that there is currently a lack of research, especially research that is empirical and comparative, about GIDEPs. In order to overcome this, a twenty-country comparative analysis was conducted to better understand the structure of these platforms, their technical components, and their governance. As there is a lack of research on this topic, this paper aimed to build initial foundational information that would enable future theorization and conceptualization on the topic of GIDEPs and in their relationship with the broader digital government literature. In doing this, this paper makes a strong contribution to the academic literature and addresses a clear gap in the literature.

Though this paper makes a number of contributions, there is a large amount of room for future research to expand on these findings. As GIDEPs are a form of digital infrastructure, and digital infrastructures play a key role in digital innovation [35], [36], future research could aim to understand the role that GIDEPs play in developing, creating, or expanding digital innovation ecosystems by opening up access to governmentally held data or in bringing different stakeholder groups into the platform's ecosystem e.g. for service delivery. A second aspect for future research could be to draw on the initial classification framework developed within this paper to explore how different structures for GIDEPs lead to better or worse outcomes (e.g. in terms of usage). Similar to this, this paper noted that that the governmental context appears to influence the design and structure of a country's GIDEP – future research could aim to explore this relationship between political structure and GIDEP structure in further detail. Additionally, while this study identified several building blocks likely to be important for future GIDEPs, it is also likely that other building blocks exist; future research could aim to further expand upon this research and identify further components of future platforms.

With this in mind, it is still possible to highlight the core contributions made by this paper, which are relevant both for academics and for practitioners. The first contribution is in its conceptualization of GIDEPs and in anchoring GIDEPs within the broader theoretical literature on digital innovation and digital infrastructure. Related to this, a second contribution of this paper is in its creation of a typology and conceptualization for GIDEPs. These are important contributions as they enable future studies to easily understand and classify GIDEPs. From a practitioner perspective, this is also an important contribution in that it easily enables a government to compare their system against others. Similar to this, an additional important contribution is related to the exploration and identification of the governance structure of GIDEPs. This is particularly of importance for practitioners who may be working to restructure, build, or reorganize how their interoperability initiatives are governed and regulated. A third, and likely most important, contribution that the paper makes is in conceptualizing the continuum of interoperability and in identifying core components likely to be present in future-oriented GIDEPs, such as the movement to API based data exchange, MyData modules, cloud based hosting, and development following a microservice architecture. Understanding the future of GIDEPs, their design, and components is essential for understanding the future of interoperability and, ultimately, in understanding how digital governments will function and innovate. This contribution will be of interest for anyone working with, either in an academic or practical sense, interoperability and data exchange platforms.

Looking toward the future it is certain that GIDEPs will play an increasingly important role in the overall digitalization of government and society. It is paramount that academics and practitioners devote the appropriate resources to understanding GIDEPs and their structures so that their full positive potential may be harnessed. If done right, GIDEPs will enable massive leaps in digital innovation, improvements in service provision, and transform how governments operate in the global information age.

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